Experiment 11: Camomile Tea — Sub-30°C Vapor and Instantaneous Motion

Experimental Protocol

Container: Stainless steel pot (leiteira). Liquid: ~500 ml filtered water with camomile, fennel, and macela (dried herbal infusion blend). Heat source: Domestic gas flame at maximum intensity, largest burner. Thermometry: Digital probe thermometer (TP300, immersed in liquid). Recording: Continuous smartphone video, frame-by-frame analysis.

Complete Data: Experiment 11 (3 Collages, 9 Frames)

Coll.	Fr	Time	Flame	T (°C)	Motion	Vapor	Observation
C1	F1	1.033	OFF	29.1	–	–	Baseline. Flame OFF
C1	F2	1.067	**ON**	29.1	–	–	Flame ignited. T unchanged. Red circle: ROI
C1	F3	6.548	ON	29.1	+	–	~5.5 s flame: motion visible. T still 29.1°C
C2	F1	10.000	ON	29.1	++	–	~9 s flame: evident motion. T still 29.1°C
C2	F2	33.533	ON	**29.0**	++	–	~32 s: T DROPS to 29.0°C. Red: flame + display
C2	F3	41.833	ON	**29.5**	++	(+)	~41 s: T rising. Vapor beginning visible
C3	F1	41.900	**OFF**	29.5	+	+	Flame OFF at 29.5°C. "OFF" label
C3	F2	45.804	OFF	**30.6**	+	++	~4 s post-OFF: T=30.6. Visible vapor (white annot.)
C3	F3	63.200	OFF	**34.4**	+	++	~21 s post-OFF: T=34.4. +4.9°C rise w/o flame

Photographic Documentation

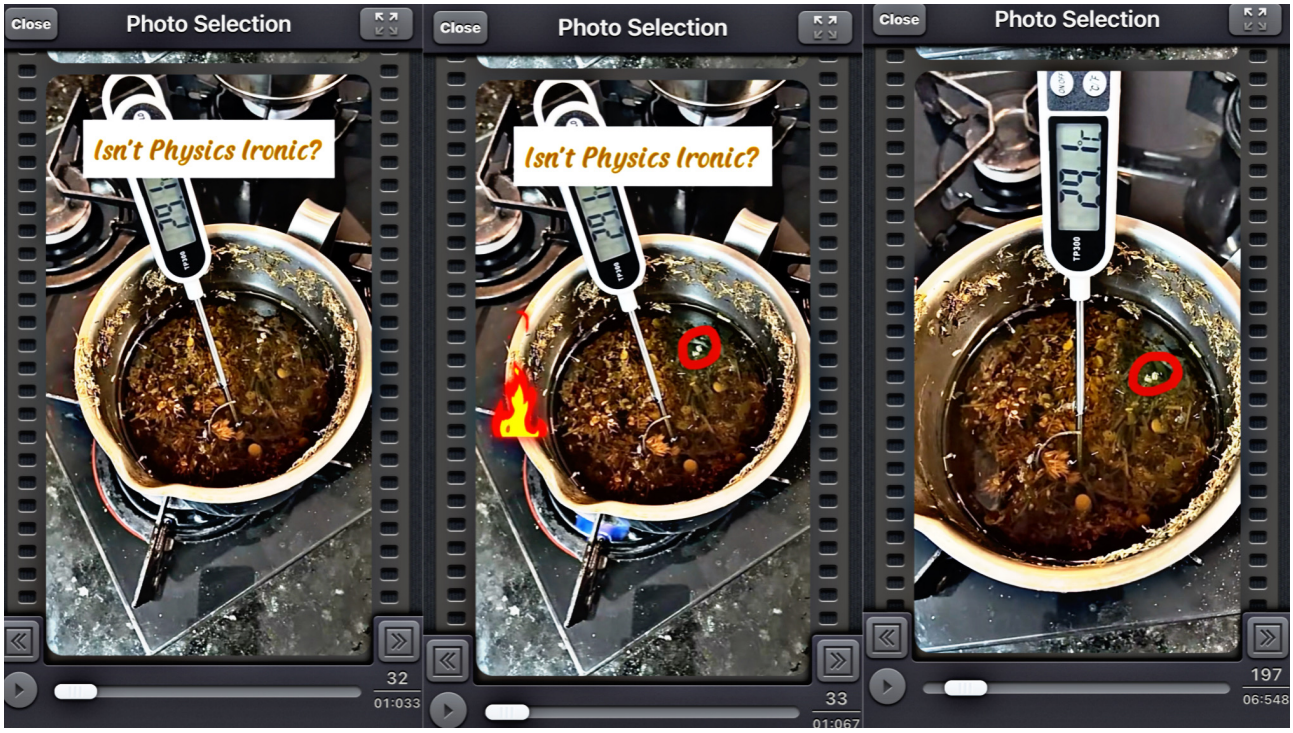

Figure 35: E11c1 — BASELINE + FLAME IGNITION + FIRST MOTION (3 frames). Stainless steel pot with camomile/fennel/macela infusion, digital probe thermometer immersed. [F1, t=1.033 s, flame OFF, T=29.1°C]: Baseline. Liquid at rest, no motion, no vapor. [F2, t=1.067 s, flame ON, T=29.1°C]: Flame ignited. Temperature unchanged. Red circle marks surface region of interest for motion observation. [F3, t=6.548 s, flame ON, T=29.1°C]: **FIRST VISIBLE MOTION** — ~5.5 seconds after ignition, surface motion visible to naked eye. Temperature has NOT changed. Motion begins while temperature is perfectly stable. The thermal force field activates before the thermometer registers any energy input.

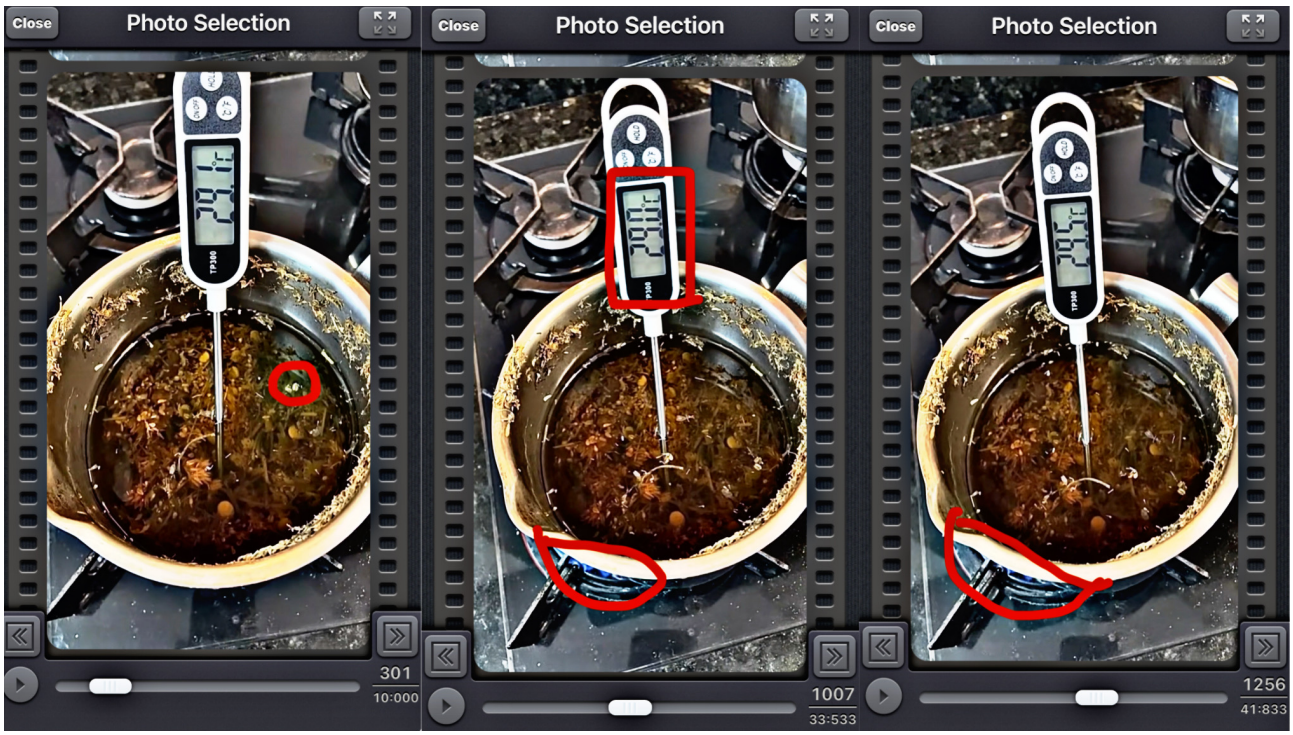

Figure 36: E11c2 — SUSTAINED MOTION + TEMPERATURE DROP + VAPOR ONSET (3 frames). [F1, t=10.000 s, flame ON, T=29.1°C]: ~9 seconds of flame. Motion evident (++). Temperature still 29.1°C — the system absorbs all incoming energy endothermically. Red circle marks surface. [F2, t=33.533 s, flame ON, T=**29.0°C**]: **TEMPERATURE DROPS** after ~32 seconds of maximum flame. Red annotations mark thermometer display (29.0) and flame below pot. This reproduces the Experiment 5 pattern: heating DECREASES temperature. [F3, t=41.833 s, flame ON, T=29.5°C]: Temperature recovering. Vapor becoming visible at sub-30°C — visible in video at the exact moment temperature begins its definitive ascent, when internal energy storage capacity is exhausted and incoming energy converts to mechanical work + molecular agitation.

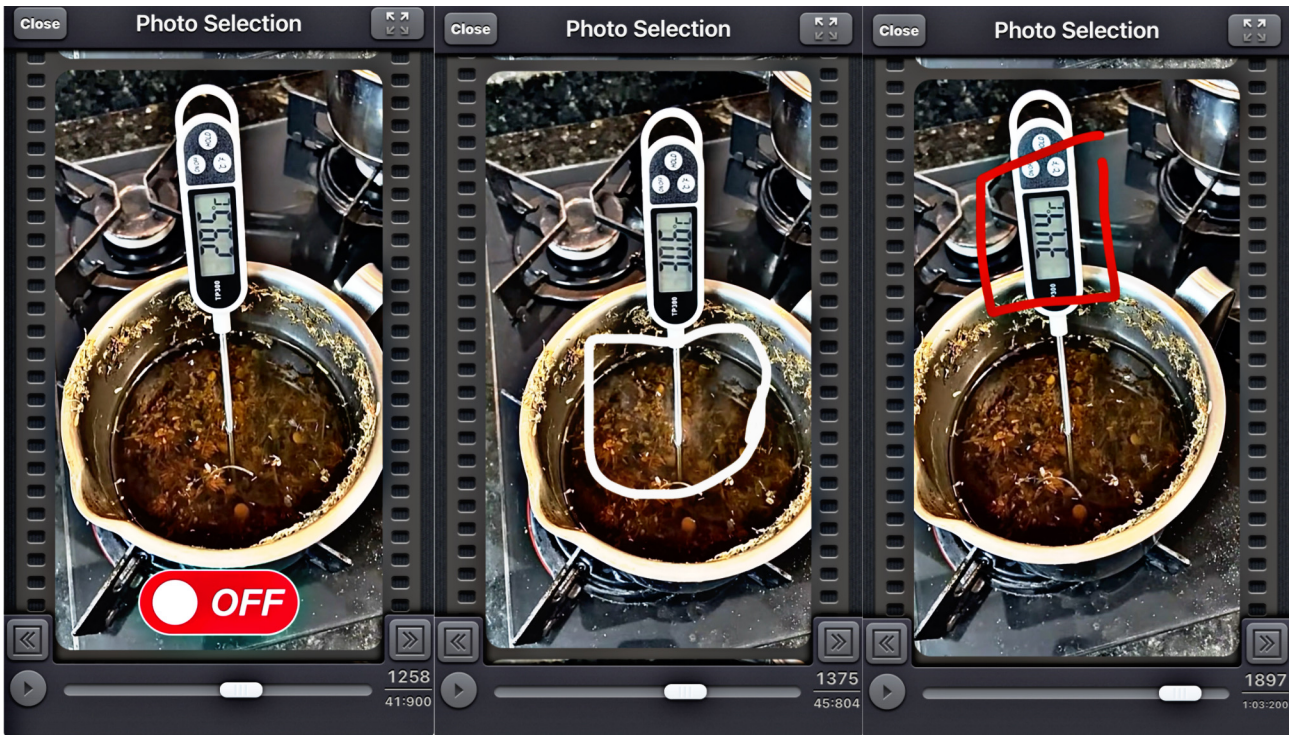

Figure 37: E11c3 — FLAME OFF + THERMAL RESERVOIR + VAPOR INTENSIFICATION (3 frames). [F1, t=41.900 s, flame OFF, T=29.5°C]: Flame extinguished. "OFF" label visible. After ~41 s of maximum flame, temperature rose only 0.4°C from baseline. [F2, t=45.804 s, flame OFF, T=**30.6°C**]: ~4 s after flame OFF — temperature jumps +1.1°C. White annotation marks visible vapor, now clearly evident. Vapor INCREASES after flame OFF. Stored internal energy is released as the system equilibrates. [F3, t=63.200 s, flame OFF, T=**34.4°C**]: ~21 s after flame OFF — +4.9°C rise without flame (29.5→34.4°C). **THERMAL RESERVOIR CONFIRMED** in a third solute system.

Anomaly Analysis — Experiment 11

Anomaly H1: Visible bulk motion at constant temperature. Surface motion becomes visible within ~5.5 seconds of flame ignition (C1 F3) while temperature remains at exactly 29.1°C. Motion is evident and continuous by t=10 s (C2 F1), still at 29.1°C. Classical convection requires a temperature differential; none exists. The thermal force field drives motion independently of stored thermal energy.

Anomaly H2: Temperature decreases during sustained heating. After ~32 seconds of maximum flame, temperature drops 29.1→29.0°C (C2 F2). Reproduces Experiment 5 (–1.5°C in 41 s) and Experiment 6 (46-second plateau). Consistency across legumes, herbs, and pure water confirms this is a general property of active thermal flux interaction with matter.

Anomaly H3: Visible vapor at sub-30°C. Vapor visible while bulk temperature is below 30°C — approximately 70°C below boiling point. At 29.5°C bulk with ambient ~25°C, the $\delta T \approx 4.5$°C is far below levels producing visible vapor from plain water. The active thermal flux creates a phase-change field producing vapor at thermodynamically insufficient temperatures.

Anomaly H4: Temperature rises +4.9°C in ~21 s without flame. From 29.5°C to 34.4°C after flame extinction. Post-flame heating rate (+0.23°C/s) exceeds during-flame rate (+0.01°C/s). Reproduces the thermal reservoir of Experiment 6 in a third solute system.

Systematic Falsification — Experiment 11

Classical Prediction	Observation	Status
Convection requires temperature differential	Visible motion at constant 29.1°C within 5.5 s	**FALSIFIED**
Heating increases temperature monotonically	T drops 29.1→29.0°C after 32 s of max flame	**FALSIFIED**
Visible vapor requires T well above ambient	Vapor at 29.5°C bulk, ~4.5°C above ambient	**FALSIFIED**
T decreases when heat source removed	T rises +4.9°C in 21 s after flame OFF	**FALSIFIED**
Heating rate maximal during active flame	Post-flame +0.23°C/s vs during-flame +0.01°C/s	**FALSIFIED**

Cross-Experiment Consistency

Experiment 11 reproduces four anomaly patterns in a third solute system (herbal infusion, after beans/lentils in Exp. 5–6 and pure water in Exp. 3–4):

Pattern	First Documented	Exp. 11
Motion before T change	Exp. 3: motion at 72.7°C, still at 100°C	Motion at constant 29.1°C in 5.5 s
T drop during heating	Exp. 5: –1.5°C in 41 s	–0.1°C in 32 s
Temperature plateau	Exp. 6: 46 s at 22.0°C	~30 s at 29.1°C
Post-flame T rise	Exp. 6: +11.7°C in 17 min 45 s	+4.9°C in 21 s

The universality across three distinct solute systems eliminates material-specific explanations and confirms that the anomalies are properties of the thermal force field interaction with solute-laden water under active flux.

References

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Supplementary materials: Google Drive · YouTube: @5incere5science · TikTok: @5incere5science

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